TUT-7

22/10/20

① Consider a large (radius of 25 m) iron-nickel meteorite collides with earth at an estimated speed of 1.28×10^4 m/s. If the impact is lasted for 2 sec and iron-nickel meteorite has a density (ρ) of 7970 kg/m³, estimate the average force the meteor applied to earth during the impact.

impulse

⇒ we know, $I = (\text{Fava}) \times \Delta t$

$\Rightarrow \text{Fava} = \frac{I}{\Delta t}$

we know, impulse, I = change in momentum.

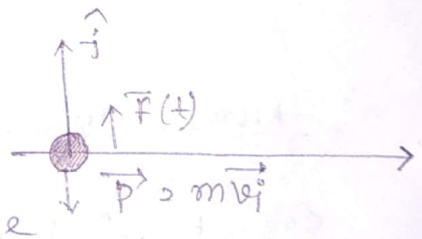
$= m\Delta v$ [assuming, m as const]

$\therefore \text{Fava} = \frac{m\Delta v}{\Delta t}$

$= \frac{\rho(\frac{1}{3}\pi r^3).(v_f - v_i)}{\Delta t}$

$m = \rho V$
$= (\frac{4}{3}\pi r^3)\rho$

given, $\Delta t = 2$ sec.



Considering final velocity to be zero,

$$\text{Fava} = \frac{(7970\ \text{kg/m}^3)(\frac{4}{3}\pi(25\text{m})^3)[0 - (-1.28\times10^4\hat{j}\ \text{m/s})]}{2\ \text{sec}}$$

(Note that, the impulse is along $\hat{j}$ dirn but initial velocity was along $-\hat{j}$.)

[force applied by earth on the meteorite : $\hat{j}$]

$\therefore \boxed{\text{Fava} = 3.33\times10^{12}\ \text{N}}$

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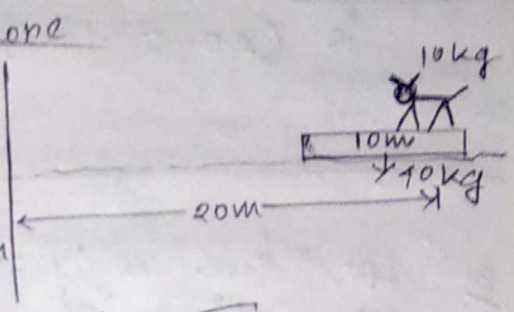
Q. A dog of mass 10 kg is standing on a flat 10m long boat so that is 20 meters from the shore. It walks 8m on the boat towards the shore and the stops. The mass of the boat is 40 kg and friction between the boat and the water surface is negligible. How far is the dog from the shore now?

⇒



Consider the boat and the dog as a system.

Initially, COM of the system is at rest.

Since no external force is acting on the system, hence COM of system will remain stationary.

Let, initially distance of COM of the boat from the shore be 'x m'.

So, x_{1com} = distance of the COM of the system from the shore

$$= \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

$$= \frac{40x + 10 \times 20}{(40+10)} \text{ m}$$

$$= \frac{40x + 200}{50} \text{ m} \quad \text{—(1)}$$

Since, the dog moves towards the shore, for the COM of the system to be at rest. The boat has to move away from the shore.

Let, distance moved by the boat be x'.

Then $x_{2com} = \frac{40(x+x') + 10(20 - 8 \times x')}{(40+10)}$ m —(2)

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Now, $x_{1\,com} = x_{2\,com}$

$$\Rightarrow \frac{40x - 1200}{\cancel{50}} = \frac{40(x+x') + 10(12+x')}{\cancel{50}}$$

$$\Rightarrow \boxed{x' = 1.6\,m}$$

Hence, distance of the dog from the stone is $(20 - 8 + 1.6)\,m = 13.6\,m$.

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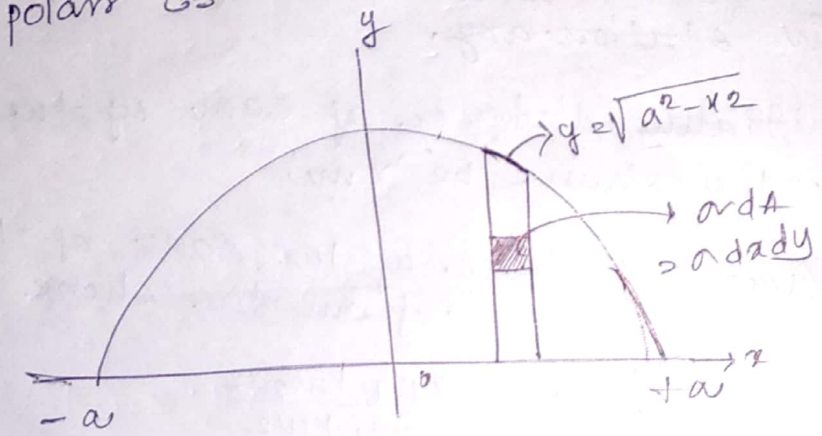
3. Find the centroid of a semi-circular region of radius 'a' using (a) Cartesian co-ordinate system.

(b) polar c.s.

$\Rightarrow$



eqn of the semi-circle $\rightarrow$ $x^2 + y^2 = a^2$

$\Rightarrow y = \sqrt{a^2 - x^2}$ $[\because y \geq 0]$

If, σ is the mass per unit area, assumed constant, then the co-ordinates of the centroid is given by,

$$x = \frac{\int x\, dm}{\int dm} = \frac{\int x \sigma\, dA}{\int \sigma\, dA} = \frac{\iint x\, dx\, dy}{\iint dx\, dy}$$

$$= \frac{\int_{x=-a}^{+a} \int_{y=0}^{\sqrt{a^2-x^2}} x\, dx\, dy}{\int_{x=-a}^{+a} \int_{y=0}^{\sqrt{a^2-x^2}} dx\, dy} = 0.$$

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and $\quad y = \dfrac{\int y\,dm}{\int dm}$

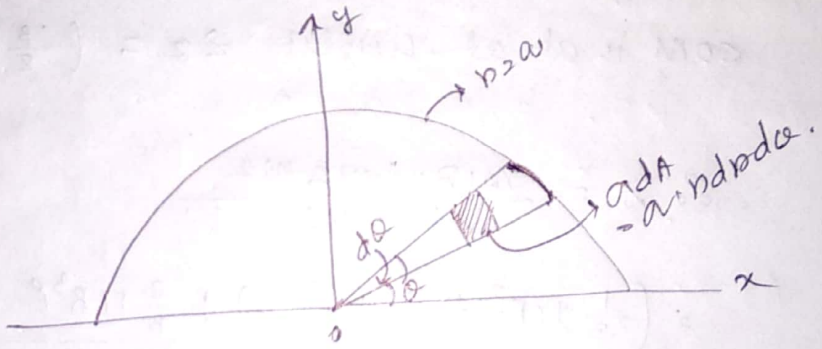
$$= \frac{\iint y\,dx\,dy}{\iint dx\,dy} = \frac{\int_{x=-a}^{+a}\int_{y=0}^{\sqrt{a^2-x^2}} y\,dx\,dy}{\int_{-a}^{+a}\int_{0}^{\sqrt{a^2-x^2}} dx\,dy}$$

$$= \frac{2a^3/3}{\pi a^2/2}$$ → area of semi-circle.

$$\boxed{y = \frac{4a}{3\pi}}$$

NB! we can easily write $\boxed{x = 0}$, by the symmetry of the centroid on the y axis.

(b)



eqn of semi-circle $r = a$

By, symmetry, the centroid must lie on the y axis. so, $x = 0$

Now By polar CS $\quad y = r\sin\theta$

$$\therefore \bar{y} = \frac{\int y\,dA}{\int dA} = \frac{\int_{\theta=0}^{\pi}\int_{r=0}^{a}(r\sin\theta)\,r\,dr\,d\theta}{\int_{\theta=0}^{\pi}\int_{r=0}^{a} r\,dr\,d\theta}$$

$$= \frac{4a}{3\pi}$$

Advantage of using polar CS instead of Cartesian CS.

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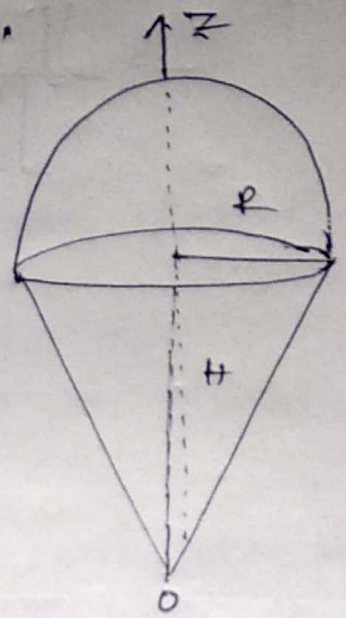
4. Find the com of a solid ~~to~~ of constant density consisting of a right circular cone of radius R and height H surrounded by a hemi-sphere of radius R.



⇒ The cm must lie on the symmetry axis of the solid.

Let, the axis be along the Z axis

The cone has a mass,

$m_1 = \left(\frac{1}{3}\pi R^2 H\right)\rho$ $[\because m = \rho V]$

its com is at a height $z_1 = \frac{3}{4}H$.

for hemi-sphere, $m_2 = \frac{2}{3}\pi R^3 \rho$.

com is at at height $z_2 = \left(\frac{3}{8}R + H\right)$ w.r.to O.

$$\therefore Z = \frac{m_1 z_1 + m_2 m_2}{m_1 + m_2}$$

$$= \frac{\left(\frac{1}{3}\pi R^2 H\rho + \frac{3}{4}H\right) + \frac{2}{3}\pi R^3 \rho \left(\frac{3}{8}R + H\right)}{\left(\frac{1}{3}\pi R^2 H\rho + \frac{2}{3}\pi R^3 \rho\right)}$$

$$= \frac{\frac{3}{4}(R^2 + H^2) + \frac{2}{3}HR}{(H + 2R)}$$

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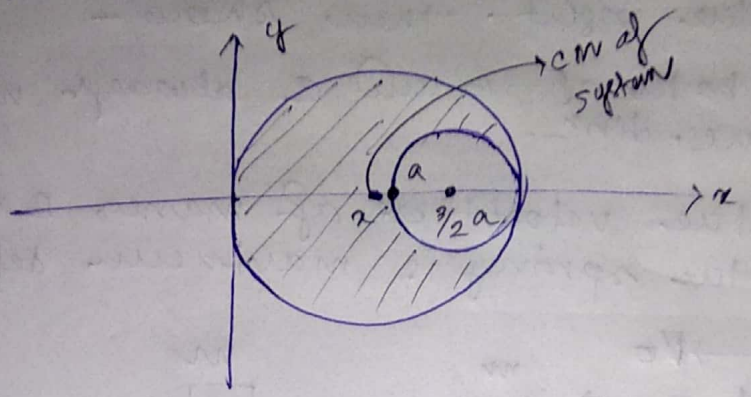
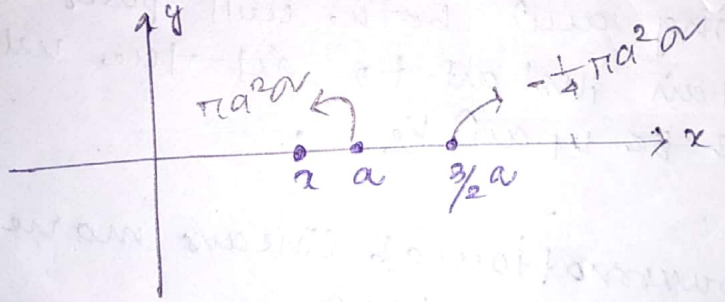
5. A circular hole of radius $a/2$ is cut out of a circular region of radius a, as shown in the figure. Find the centroid of the shaded region.

⇒



⇒ By symmetry, the centroid is located on the x axis, so that $y = 0$.

We can replace the circular ring of radius 'a' by the mass $m_1 = (\pi a^2)\sigma$. [Concentrated at its centroid at $x_1 = a$]



Illy, we can replace the circular hole of radius $a/2$ by the negative mass, $m_2 = -\frac{1}{4}\pi a^2\sigma$. Concentrated at its centroid $x_2 = \frac{3}{2}a$.

Then, the centroid is located on the x-axis at,

$$x = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

$$= \frac{(\pi a^2\sigma)(a) + (-\frac{1}{4}\pi a^2\sigma)(\frac{3}{2}a)}{\pi a^2\sigma - \frac{1}{4}\pi a^2\sigma}$$

$$\boxed{x = 5/6\, a.}$$

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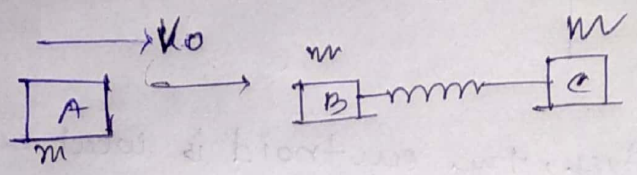
6. Two masses B and C each of value m are connected by an ideal spring and rest on a smooth surface. A third mass A of value m is given a velocity v_0 towards the right. Then show –

(i) that both of B and C always move in same dirn.

(ii) find the velocities of masses B and C when the spring is maximum deformed.

⇒



A → projectile

B → target at rest.

So, immediately after collision, the mass B has only the velocity, hence momentum and KE. But, as time passes on, energy is transferred from the body B to C with spring and both will posses velocity. At, a certain instant t, let the velocities of B and C be v_1 and v_2.

∴ By conservation of linear momentum

$$mv_0 = mv_1 + mv_2$$

By E conservation,

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 + \frac{1}{2}kx^2$$

$$v_0 = v_1 + v_2$$

$$kx^2 = v_0^2 - (v_1^2 + v_2^2)$$

$$= (v_1+v_2)^2 - (v_1^2+v_2^2)$$

$$= 2v_1v_2$$

As, kx^2 → is a positive quantity.

So, the signs of v_1 & v_2 must be same

i.e. B & C moves in same dirn.

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(ii) when the spring have the maximum deformation, the product $v_1 v_2$ ~~will~~ must be maximum i.e $v_1 = v_2$ (should be)

then, $\boxed{v_1 = v_2 = \frac{1}{2} v_0}$

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